

Open-FDD AI Agent Report

AHU · ACME Office · Detroit, MI

• June 2026, 72 samples, $\Delta t \approx 60$ min, span 72.0 h

AHU_1

Sensor checks

Checks for stuck, out-of-range, or stale sensors. Only failed checks are listed when something is wrong.

Outdoor air temperature read outside its physical range (low 35.0, high 200.0). Return air temperature and supply air temperature stayed stuck flat. Outdoor air temperature jumped suddenly and changed faster than expected.

Anomaly screening

A quick look at whether each varying trend is ordinary this month. Not an equipment fault list.

Mixed air temperature and outdoor air temperature look normal.

Executive summary

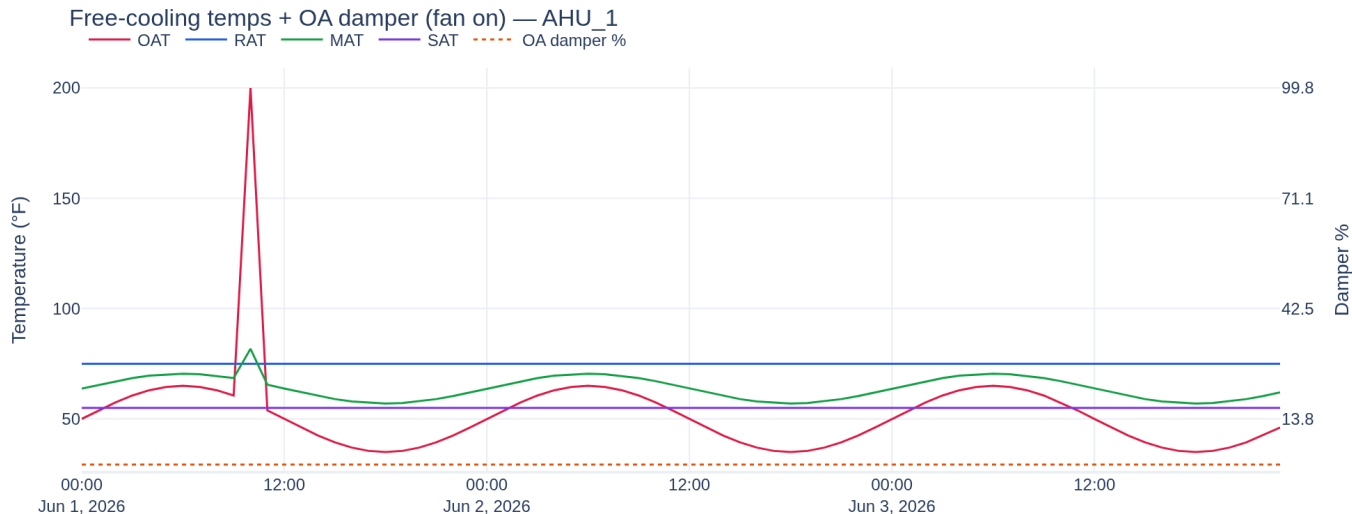
AHU_1 in June 2026 (72 samples, 72.0 h). Sensor checks: Outdoor air temperature read outside its physical range (low 35.0, high 200.0). Return air temperature and supply air temperature stayed stuck flat. Outdoor air temperature jumped suddenly and changed faster than expected. Anomaly screening: Mixed air temperature and outdoor air temperature look normal. These screening notes are not equipment faults. Confirmed operating findings: Duct static below SP at full fan (GL36 A) for 72.0 h; Economizer stuck closed for 26.0 h; Always-on fan or pump runtime for 72.0 h. The fan was on for 100.0 percent of the samples.

RCx week (2026-06-01 to 2026-06-03 UTC)

One week inside the filtered month. Figures follow the mapped roles for this system profile.

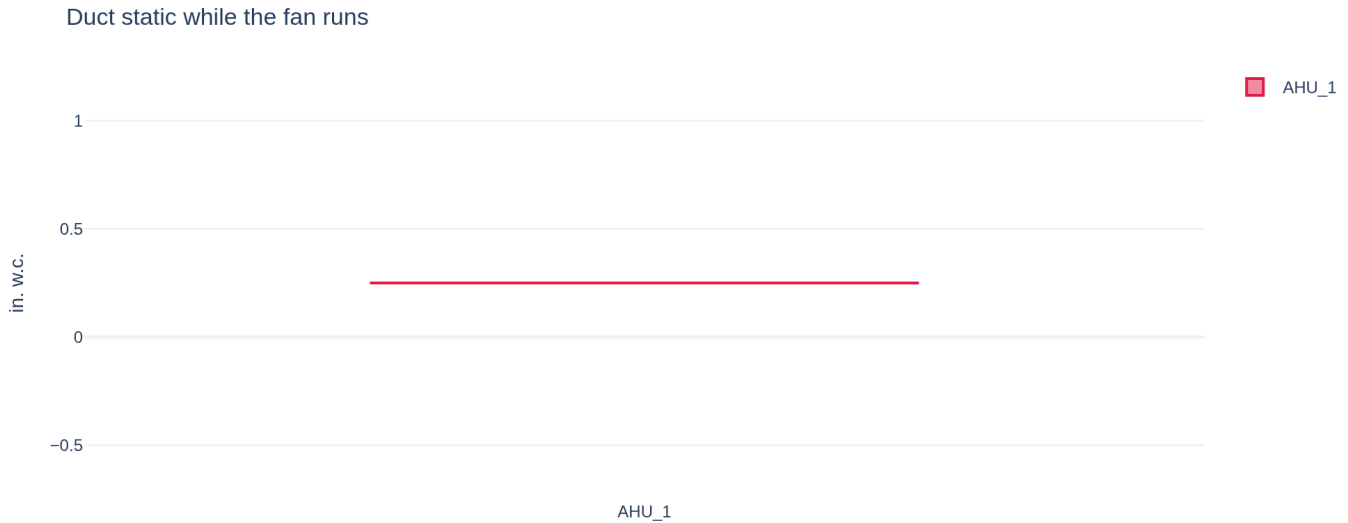
Free-cooling temperatures and outdoor-air damper

Outdoor, return, mixed, and supply air on the temperature axis. Damper percent is on the right axis. Fan running only.



Duct static while the fan runs

Duct static pressure with the fan running.



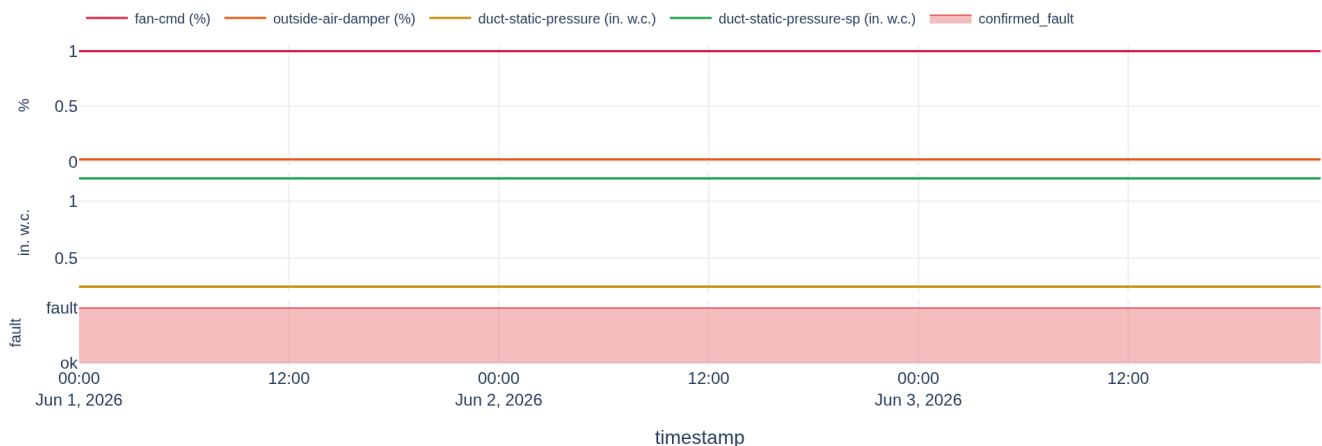
Confirmed faults

A figure is included only when the cookbook rule has confirmed fault hours in the filtered month.

FC1 – Duct static below SP at full fan (GL36 A)

Troubleshoot: Flags duct static too low while the supply fan is near full speed. Check: $DSP < DSPSP - \epsilon DSP$ AND $VFDSPD \geq 100\% - \epsilon VFDSPD$.

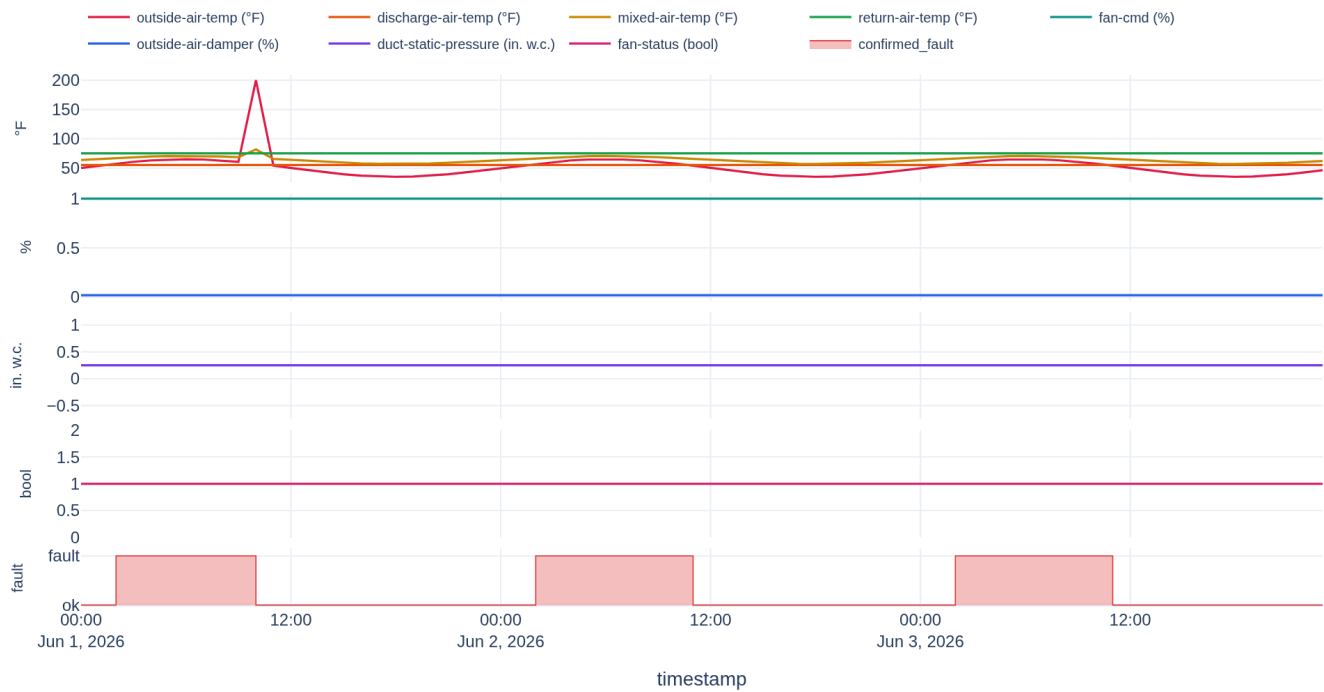
In the data: In June 2026 this holds for 72.0 hours, 2026-06-01 00:00 UTC through 2026-06-03 23:00 UTC. duct static pressure was on for 0.0 percent of those samples; duct static setpoint was on for 100.0 percent of those samples; fan speed command was on for 100.0 percent of those samples.



ECON-1 – Economizer stuck closed

Troubleshoot: Fan on, OA damper < 5%, OAT > 55°F (should be economizing). Check: Fan on, OA damper < 5%, OAT > 55°F (should be economizing).

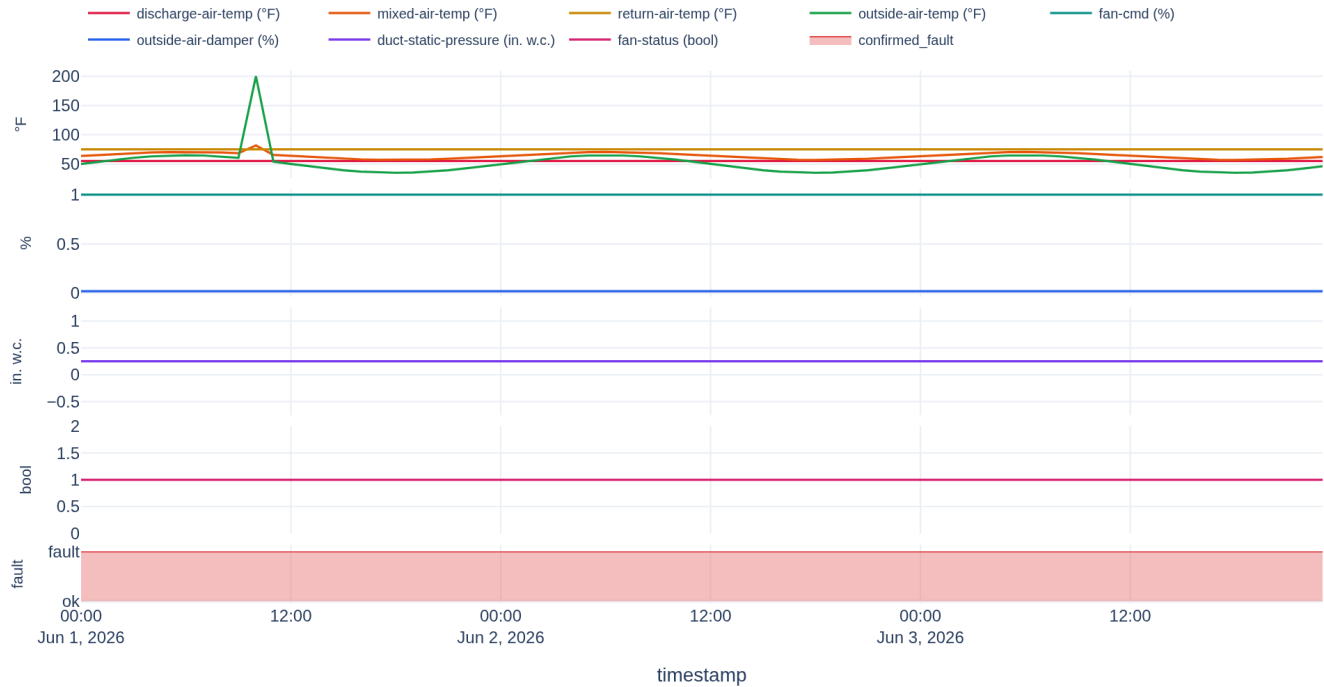
In the data: In June 2026 this holds for 26.0 hours, 2026-06-01 02:00 UTC through 2026-06-03 10:00 UTC. fan speed command was on for 100.0 percent of those samples; outdoor air damper was on for 0.0 percent of those samples; outdoor air temperature was about 63.0.



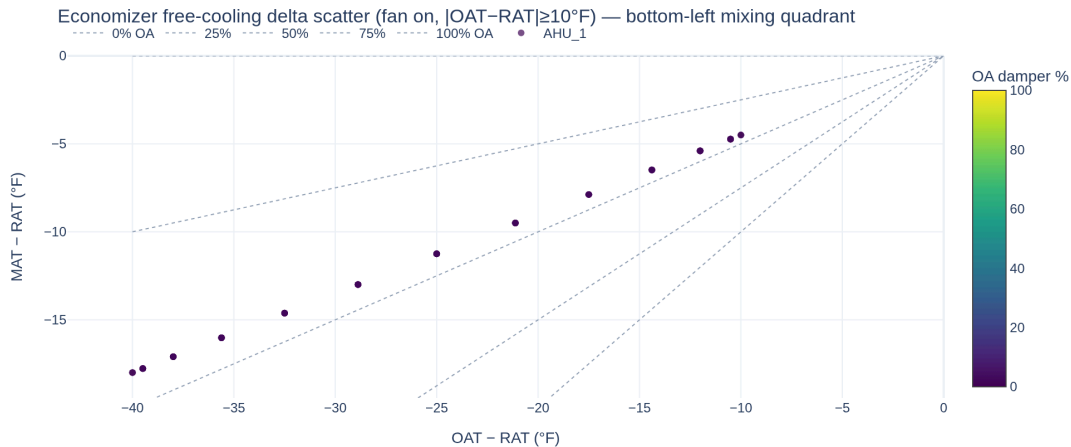
SCHED-247 – Always-on fan or pump runtime

Troubleshoot: Flags fans or pumps that stay on for nearly the entire analysis window (24/7 runtime). Check: Ranked proof (status/current > command) is on for $\geq$ always_on_pct of the window. Pressure is inferred runtime only and does not trip this rule ID.

In the data: In June 2026 this holds for 72.0 hours, 2026-06-01 00:00 UTC through 2026-06-03 23:00 UTC. fan status was on for 100.0 percent of those samples; fan speed command was on for 100.0 percent of those samples; duct static pressure was on for 0.0 percent of those samples.



Economizer delta scatter



Outdoor-air mixing with the supply fan on. The horizontal axis is outdoor-air temperature minus return-air temperature (OAT – RAT). The vertical axis is mixed-air temperature minus return-air temperature (MAT – RAT). The view is the bottom-left diagnostic quadrant only, where outdoor air is at or below return air and mixed air is at or below return air. A sample is plotted only when the fan is on and the absolute difference between outdoor air and return air is at least 10°F, so the outdoor-air fraction is identifiable.

Dotted reference lines are the outdoor-air fraction that would put mixed air on that line: 0%, 25%, 50%, 75%, and 100% outdoor air. When an outdoor-air damper is mapped, point color is damper position from closed (0%) to open (100%). In theory, points at a given outdoor-air fraction lie near the matching fraction line. A high damper position (yellow, near fully open) should cluster on the 100% outdoor-air line. Intermediate colors should sit near their own fraction lines.

Damper percent is actuator position. It is not the calculated fresh-air fraction, and it is not an airflow fraction.

Points off their fraction line indicate a mixing problem, a temperature-sensor error, or the wrong outdoor-air volume (too much or too little outdoor air).

The plot is clearest in extreme cold or hot outdoor weather, when outdoor air and return air are far apart. In mild economizer weather, where mixed air, return air, and outdoor air are nearly the same temperature, the differences shrink and those errors are hardest to see.